Math 2059 Homework 1

Deadline: 5pm 8th December 2025.

Preparation and submission: You can discuss about the solutions with your friends, but you must solve the questions on physical papers with your own handwriting. You must **submit** the homework to me **in person,** either during the lectures or by handing it to me when you find me in my office (M2 232), but the HW sent via online channels such as e-mail will be ignored and won't be considered for grading!

Note: The answers in this HW may not necessarily be Integers. For consistency and precision, please present your results using **three digits after the decimal point** if the result is not Integer.

Problem 1) Employ the following methods to find the maximum of:
$f(x) = 4x - 1.8x^2 + 1.2x^3 - 0.3x^4$

maximum: higher f(x)
minimum: lower f(x)
is xopt

a) Golden-Section search ($x_l = -2$, $x_u = 4$, $\boldsymbol{\varepsilon_s = 10\%}$). Use formulas:

$$\phi = \frac{\sqrt{5}-1}{2} \approx 0.618, \qquad d = \phi(x_u - x_l)$$

$$x_1 = x_l + d, \quad x_2 = x_u - d.$$

$$\varepsilon_a \approx \frac{1-\phi}{|x_{\text{opt}}|} |x_u - x_l| \times 100\%.$$

Where x_{opt} is whichever of x_1 or x_2 yields the better (optimal) function value.

b) Newton's method ($x_0 = 3$, $\boldsymbol{\varepsilon_s = 1\%}$)

Task 1: Watch https://youtu.be/GJODOGq7cAY
Task 2: Watch https://youtu.be/QQPz3eXXgQI
Task 3: Read Chapter 14 Multidimensional Unconstrained Optimization in the reference book.

Problem 2: Find the directional derivative of $f(x, y) = x^2 + 2y^2$ at $x = 2$ and $y = 2$ in the direction of $h = 2i + 3j$.

Problem 3: Find the gradient vector and Hessian matrix for the following function:
$f(x, y) = 2xy^2 + 3e^{(xy)}$

Problem 4: Find the minimum value of the function $f(x, y) = (x - 3)^2 + (y - 2)^2$

a) Using analytical methods. i.e. derivatives and hessian matrix etc.
b) Although this problem can be solved analytically, most practical problems cannot. In such cases we use numerical methods. Let's assume the solution of the problem is non-analytic and **solve it using the Optimal Steepest Descent method**. Refer to Example 14.4 in the book but note that in this case you are finding a minimum and using the descent, not ascent, method. Carry out only one iteration and compare the result with the analytical method.